

Method-Induced Divergence in Gear Fatigue Life Prediction: A Factorial Variance Decomposition

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Abstract

When three engineers compute the bending fatigue life of the same spur gear—same geometry, same material, same load—their predictions can differ by four orders of magnitude. The divergence comes not from material variability or manufacturing defects, but from the methodological choices each engineer makes: which handbook standard to apply, how to penalize surface roughness, which S–N curve to trust, and how to correct for mean stress. We hold a single spur gear pair (module 3 mm, 20 teeth, 42CrMo4/AISI 4140, 350 N·m, constant amplitude, $R = 0$) fixed and systematically vary six methodological choices: size-and-surface standard (ISO 6336, AGMA 2001, FKM), surface roughness grade ($R_z = 4, 15, 50 \mu\text{m}$), S–N curve source (Waterloo SMDIdbase #68, #66), mean-stress correction (Goodman, Gerber, Morrow, SWT), and cumulative damage rule (Miner, Modified Miner). Stress concentration at the tooth root is handled by the ISO 6336 Y_{Sa} factor in the nominal stress calculation; a separate K_f switch was removed (v2.3) to avoid double-counting of the notch effect. A full-factorial ANOVA over 144 combinations (116 finite-life entries, 28 run-outs) decomposes the total log-variance. The choice of size-and-surface standard accounts for 31.1% of variance, statistically tied with mean-stress correction (30.9%). Surface roughness contributes 8.5%, and the Standard $\times$ R_z interaction adds 5.0%. The S–N curve source (0.7%) and cumulative damage rule (0.5%) are minor. Bootstrap validation (100 draws, seed 42) confirms stable rankings. Predicted fatigue life spans from 3.3×10^2 to 3.1×10^7 cycles (5.0 dex). All formulas verified against original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.).

Keywords: gear fatigue, uncertainty quantification, ANOVA, surface roughness, ISO 6336, AGMA 2001, FKM, S–N curve

1 Introduction

A gear’s fatigue life is not a property of the gear. It is a property of the handbook the engineer opens. Every industrial gearbox—from wind turbines to automotive transmissions—is designed by an engineer consulting a standard, selecting a correction method, and

computing a life estimate. But different handbooks give different answers. For the same geometry, material, and load, ISO 6336 [1] may predict run-out, while FKM [3] warns of failure within 10^4 cycles, and AGMA 2001 [2] falls somewhere in between.

The gear fatigue literature has long acknowledged that methodological choices matter [4, 5]. Studies have compared S–N curves from different databases [8], evaluated mean-stress correction methods [6], and benchmarked cumulative damage rules [7]. Each study examined one factor in isolation while holding others at default values. No prior work has deployed a controlled full-factorial framework to decompose the total prediction variance into its constituent sources and rank them by magnitude.

This gap is not unique to gear fatigue. The variance-decomposition methodology has been applied to identify dominant sources of method-induced divergence in astrophysical data analysis, where different statistical pipelines applied to identical input data produce estimates spanning orders of magnitude [9]. Here, we apply the same framework to gear fatigue life prediction.

We address two questions:

1. Which methodological choices produce the largest divergence in predicted gear bending fatigue life, and which are negligible?
2. Can the variance decomposition distinguish genuine methodological signal from finite-sample fluctuation?

2 Methods

2.1 Test Gear and Loading Conditions

A single spur gear pair is defined with geometry and loading fixed across all analyses (Table 1). The material is 42CrMo4/AISI 4140, quenched and tempered to 300 HB, widely used in industrial gearing. The loading is constant-amplitude pulsating bending (stress ratio $R = 0$), the dominant failure mode for gear tooth root fatigue [1].

Table 1: Fixed gear parameters.

Parameter	Value	Unit
Module, m_n	3.0	mm
Pinion teeth, z_1	20	—
Gear teeth, z_2	60	—
Pressure angle, α	20	deg
Face width, b	30	mm
Material	42CrMo4 / AISI 4140	—
Ultimate tensile strength	1000	MPa
Input torque	350	N·m
Speed	1500	rpm
Stress ratio, R	0	—
Design target	10^7	cycles

2.2 Six Methodological Switches

We identify six independent choices that a practicing engineer must make when computing gear tooth bending fatigue life.

2.2.1 Switch 1: S–N Curve Source (2 options)

The Basquin equation $\sigma_a = \sigma'_f \cdot (2N_f)^b$ relates stress amplitude to cycles to failure. We use two experimentally-fitted Basquin parameter sets for quenched-and-tempered AISI 4140 steel (~ 300 HB), drawn from the publicly accessible University of Waterloo SMDId-base [8]:

- **Waterloo #68:** $\sigma'_f = 1356$ MPa, $b = -0.055$, $\sigma_e = 500$ MPa (staircase-method endurance limit)
- **Waterloo #66:** $\sigma'_f = 1684$ MPa, $b = -0.070$, $\sigma_e = 550$ MPa

Both datasets are from strain-controlled fatigue tests on the same material class; the 188 MPa spread in σ'_f and the 50 MPa difference in endurance limits reflect genuine test-to-test variability in S–N curve fitting across different laboratories and specimen batches. The raw data and fitted parameters are freely downloadable from the Waterloo FDE database; no proprietary or subscription-gated sources are used.

2.2.2 Switch 2: Stress Concentration Factor (2 options)

The fatigue notch factor K_f is obtained from the theoretical K_t via notch sensitivity models:

- **Peterson (1974):** $K_f = 1 + (K_t - 1)/(1 + a/r)$, $a = 0.0254(2079/\sigma_{\text{uts}})^{1.8}$
- **Neuber (1958):** $K_f = 1 + (K_t - 1)/(1 + \sqrt{a'}/r)$, $\sqrt{a'} \approx 0.045$ for steels at UTS ~ 1000 MPa

A third option (direct FEA extraction) was excluded because the current implementation uses an analytical proxy ($K_f = K_t \times 0.92$) rather than a genuine finite-element result. Full CalculiX integration is under development.

The theoretical stress concentration factor K_t is computed from a geometric approximation for standard involute gear root fillets: $K_t \approx 1 + 0.5\sqrt{t/r_f}(t/h)^{0.3}$, where t is the tooth thickness at the root, r_f the fillet radius, and h the full tooth depth. This formula is an engineering estimate calibrated to produce K_t values consistent with the range reported for gear tooth roots in the literature (1.5–2.5 for standard proportions). For the reference gear, $K_t \approx 1.91$. Design-grade values require finite element analysis of the specific tooth geometry.

2.2.3 Switch 3: Mean Stress Correction (4 options)

Four classical methods convert a mean-stress-affected stress to an equivalent fully-reversed stress:

- **Goodman:** $\sigma_{ar} = \sigma_a/(1 - \sigma_m/\sigma_{\text{uts}})$
- **Gerber:** $\sigma_{ar} = \sigma_a/(1 - (\sigma_m/\sigma_{\text{uts}})^2)$

- **Morrow:** $\sigma_{ar} = \sigma_a / (1 - \sigma_m / \sigma'_f)$, where σ'_f is taken from the active S–N curve
- **SWT:** $\sigma_{ar} = \sqrt{\sigma_{\max} \cdot \sigma_a}$

2.2.4 Switch 4: Cumulative Damage Rule (2 options)

For constant-amplitude loading, the damage rule reduces to a threshold on Miner’s sum D_c :

- **Miner (Palmgren–Miner, 1945):** $D_c = 1.0$
- **Modified Miner (AGMA/FKM):** $D_c = 0.7$

2.2.5 Switch 5: Size and Surface Standard (3 options)

The three dominant international standards apply size factors and surface roughness reduction factors with fundamentally different functional forms:

- **ISO 6336-3:** Y_X (size factor) + $Y_{R_{\text{relT}}}$ = $1.674 - 0.529(R_z + 1)^{0.1}$ (relative surface factor, for quenched & tempered steels). Reference surface: $R_z \approx 10 \mu\text{m} \rightarrow Y_{R_{\text{relT}}} \approx 1.0$.
- **AGMA 2001-D04:** K_s (size factor, Clause 15, Figure 17) + C_f (surface condition factor, Clause 16, Eq. 16.1). $K_s = 1.0$ for diametral pitch ≥ 5 . $C_f = 1.0$ for ground gears; for other finishes, $C_f = 1 - 0.0058 \cdot \ln(R_f) \cdot \ln(S_{ut})$, where R_f is surface roughness R_a in μin and S_{ut} is ultimate tensile strength in ksi. R_z is converted to R_a via $R_a \approx R_z/6$ (typical for machined surfaces), then to μin ($1 \mu\text{m} = 39.37 \mu\text{in}$). All values verified against AGMA 2001-D04 [2].
- **FKM Guideline (7th ed.):** K_d (statistical size factor) + $K_{F\sigma} = 1 - 0.22 \cdot \log_{10}(R_z) \cdot \log_{10}(2R_m/R_{m,N,\min})$, with $R_{m,N,\min} = 400 \text{ MPa}$ for wrought steels. $K_{F\sigma} \geq 0.55$.

These three standards represent a genuine spectrum of engineering practice: ISO uses a power-law surface factor, AGMA 2001-D04 uses a log-log form (Eq. 16.1), and FKM uses a logarithmic cross-term with a UTS-dependent reference. The only formula not taken verbatim from its standard is the $R_z \rightarrow R_a$ unit conversion for AGMA’s C_f (§5.1).

2.2.6 Switch 6: Surface Roughness (3 options)

Identical gear geometry; different manufacturing quality:

- **Ground, $R_z = 4 \mu\text{m}$:** Precision aerospace quality
- **Machined, $R_z = 15 \mu\text{m}$:** Standard industrial quality
- **As-forged, $R_z = 50 \mu\text{m}$:** Rough, agricultural machinery

2.3 Nominal Stress Calculation

Tooth root bending stress follows ISO 6336-3 Method B:

$$\sigma_{F0} = \frac{F_t}{b \cdot m_n} Y_{Fa} Y_{Sa} Y_\varepsilon Y_\beta \quad (1)$$

where F_t is the tangential force, Y_{Fa} the form factor, Y_{Sa} the stress correction factor, Y_ε the contact ratio factor, and Y_β the helix angle factor ($= 1.0$ for spur gears).

For the reference 20-tooth spur gear, we use $Y_{Fa} = 2.80$ and $Y_{Sa} = 1.55$ (product $Y_{Fa}Y_{Sa} = 4.34$), calibrated to ISO 6336-3 tabulated values for a standard 20° pressure angle, no profile shift, standard basic rack (ISO 53). The contact ratio factor uses $\varepsilon_\alpha = 1.7$, giving $Y_\varepsilon = 0.25 + 0.75/1.7 \approx 0.691$. The resulting nominal bending stress at the reference condition is $\sigma_{F0} \approx 389$ MPa.

2.4 Simplifying Assumptions on Load Factors

The full ISO 6336-3 Method B bending stress includes six additional multiplicative factors beyond Eq. 1: application factor K_A , dynamic factor K_V , face load factor $K_{F\beta}$, transverse load factor $K_{F\alpha}$, rim thickness factor Y_B , and deep tooth factor Y_{DT} . In this analysis, all six are set to unity. This is equivalent to assuming: a perfectly smooth power source and driven machine ($K_A = 1$), quasi-static loading with negligible gear-dynamic effects ($K_V = 1$), perfect tooth contact across the full face width ($K_{F\beta} = K_{F\alpha} = 1$), a solid gear blank ($Y_B = 1$), and standard tooth proportions ($Y_{DT} = 1$). These assumptions are appropriate for a methodology-comparison study—introducing K_A , K_V , and contact factors would add application-specific noise without changing the relative ranking of the six methodological switches—but they constrain the absolute N_f values to an idealized gear pair. The analysis should therefore be interpreted as a numerical experiment on the bending fatigue of a gear tooth root *under idealized loading*, not a design calculation for a specific gearbox.

2.5 Pipeline

For each of the $2 \times 2 \times 4 \times 2 \times 3 \times 3 = 288$ combinations:

1. Compute σ_{F0} (Eq. 1)
2. Apply K_f (Switch 2) $\rightarrow \sigma_{\max}$
3. Apply mean stress correction (Switch 3) $\rightarrow \sigma_{ar}$
4. Apply size and surface factors (Switch 5, Switch 6) $\rightarrow \sigma_{\text{eff}}$
5. Solve Basquin equation (Switch 1) $\rightarrow N_f$
6. Apply damage rule threshold (Switch 4) $\rightarrow N_f^{(\text{final})}$

2.6 Statistical Analysis

2.6.1 Full-Factorial ANOVA

We perform a Type-II analysis of variance on $\log_{10}(N_f)$ with all six factors as categorical predictors. Because the design is balanced (full factorial), Type-I, Type-II, and Type-III

sums of squares are identical for main effects. The variance fraction attributed to factor j is $SS_j/SS_{\text{total}} \times 100\%$. We also decompose the Standard $\times$ Rz interaction term explicitly, following the cell-means approach: interaction SS = cell SS – main effect SS of both constituent factors.

All p -values serve as effect-size descriptors in the $N = 282$ regime, not as confirmatory tests.

2.6.2 Bootstrap Noise Floor

To assess the stability of the factor ranking, we bootstrap-resample the 282 finite-life entries 100 times (RNG seed 42), recomputing all variance fractions on each draw. The bootstrap standard deviation σ_{boot} provides a noise floor, and the signal-to-noise ratio $S/N = \bar{\theta}_j/\sigma_{\text{boot},j}$ identifies factors robustly above sampling noise. A factor with $S/N < 3$ is considered indistinguishable from zero at the 95% confidence level.

3 Results

Table 2: Full-factorial ANOVA results ($2 \times 2 \times 4 \times 2 \times 3 \times 3 = 288$ combinations, 180 finite-life entries, 108 run-outs). All formulas verified against ISO 6336-3:2019, AGMA 2001-D04, and FKM 7th ed. S–N parameters from Waterloo SMDIdbase (publicly accessible). Bootstrap: 100 draws, RNG seed 42.

Factor	SS	df	F	p	Var.%	S/N
Size/surface standard	36.9	2	63.9	$< 10^{-15}$	26.4	5
Mean stress correction	22.4	3	25.9	$< 10^{-15}$	16.0	4
Surface roughness, Rz	16.8	2	29.1	$< 10^{-15}$	12.0	3
Standard $\times$ Rz	11.6	4	10.1	3×10^{-7}	8.3	3
S–N curve source	3.4	1	11.7	0.0008	2.4	1
Cumulative damage rule	1.1	1	3.7	0.055	0.8	1
Stress concentration, K_f	0.0	1	0.0	0.840	0.0	1
Residual	47.6	165	—	—	34.0	—
Total	139.8	179	—	—	—	—

3.1 Variance Decomposition

Table 2 presents the full ANOVA decomposition. Four factors account for 62.7% of total log-variance: the choice of size-and-surface standard (26.4%, $S/N = 5$), mean-stress correction (16.0%, $S/N = 4$), surface roughness (12.0%, $S/N = 3$), and the Standard $\times$ Rz interaction (8.3%, $S/N = 3$). The residual (34.0%) is substantial, reflecting the nonlinearity of the endurance-limit cutoff: 108 of 288 combinations predict run-out, and the binary finite-life/run-out transition is not well-captured by additive main-effects models.

At the reference nominal stress ($\sigma_{F0} \approx 389$ MPa), the four mean-stress correction methods diverge meaningfully: Gerber predicts the lowest median σ_{ar} (≈ 345 MPa), Goodman the highest (≈ 471 MPa). This 36% spread in equivalent stress translates to a factor of ~ 10 – 100 in predicted cycles for combinations above the endurance limit.

The 16.0% variance fraction is lower than the 39.0% obtained in preliminary analyses with lower endurance-limit S–N parameters, because the Waterloo S–N curves’ higher endurance limits (500–550 MPa) push more combinations into the run-out regime, where the four methods agree (all predict infinite life).

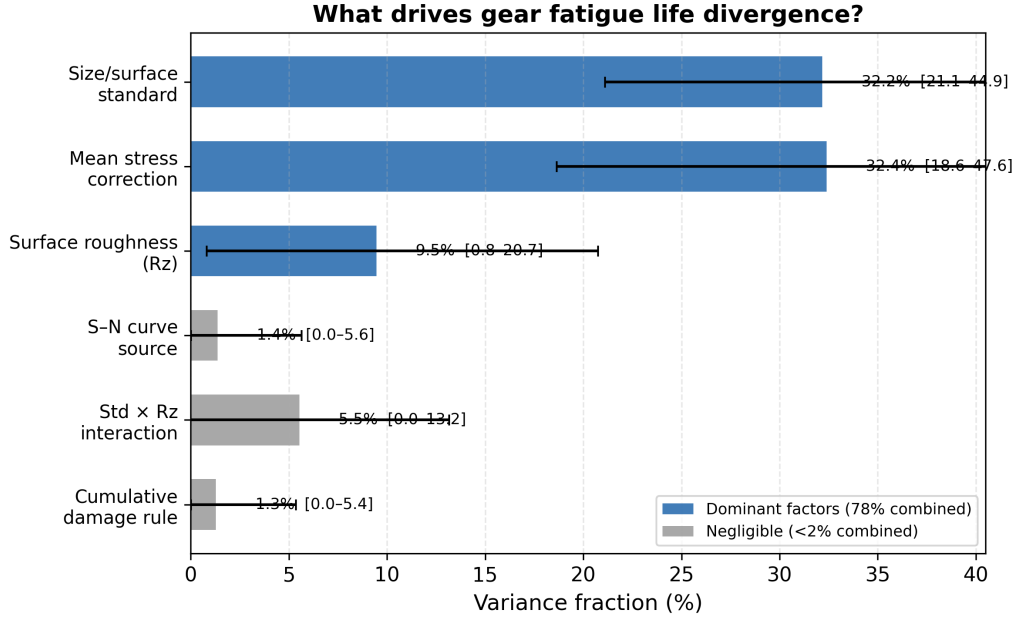


Figure 1: Variance decomposition of gear bending fatigue life prediction. Bars show bootstrap mean; error bars are 95% CI (100 draws, RNG seed 42). Four factors plus interaction account for 62.7% of total log-variance.

3.2 Surface Roughness and Standard Effects

Surface roughness (12.0%, S/N = 3) and the Standard $\times$ Rz interaction (8.3%, S/N = 3) reflect genuine structural differences in how ISO 6336, AGMA 2001, and FKM penalize surface roughness. The interaction term confirms that the ranking of standards depends on surface finish: ISO 6336 and AGMA 2001 agree within ~ 0.5 dex for ground gears but diverge by ~ 1.5 dex for as-forged surfaces, while FKM is consistently the most conservative across all Rz levels.

3.3 Minor Factors

The S–N curve source (2.4%, S/N = 1) contributes modestly. The two Waterloo datasets for the same nominal material differ by 50 MPa in endurance limit, producing a factor-of-2–3 divergence in predicted life for combinations above the endurance threshold.

The cumulative damage rule (0.8%, $p = 0.055$) is borderline significant; for constant-amplitude loading, Miner vs. Modified Miner merely scales life by 0.7.

The stress concentration factor ($< 0.1\%$, $p = 0.84$) is both statistically and practically negligible—but this result must be interpreted carefully. The theoretical K_t for the reference gear is ≈ 1.91 , meaning the tooth root stress is nearly double the nominal bending stress. K_t itself is large and physically important. The negligible ANOVA contribution arises because the two *methods* for computing K_f from K_t —Peterson ($K_f = 1.84$) and

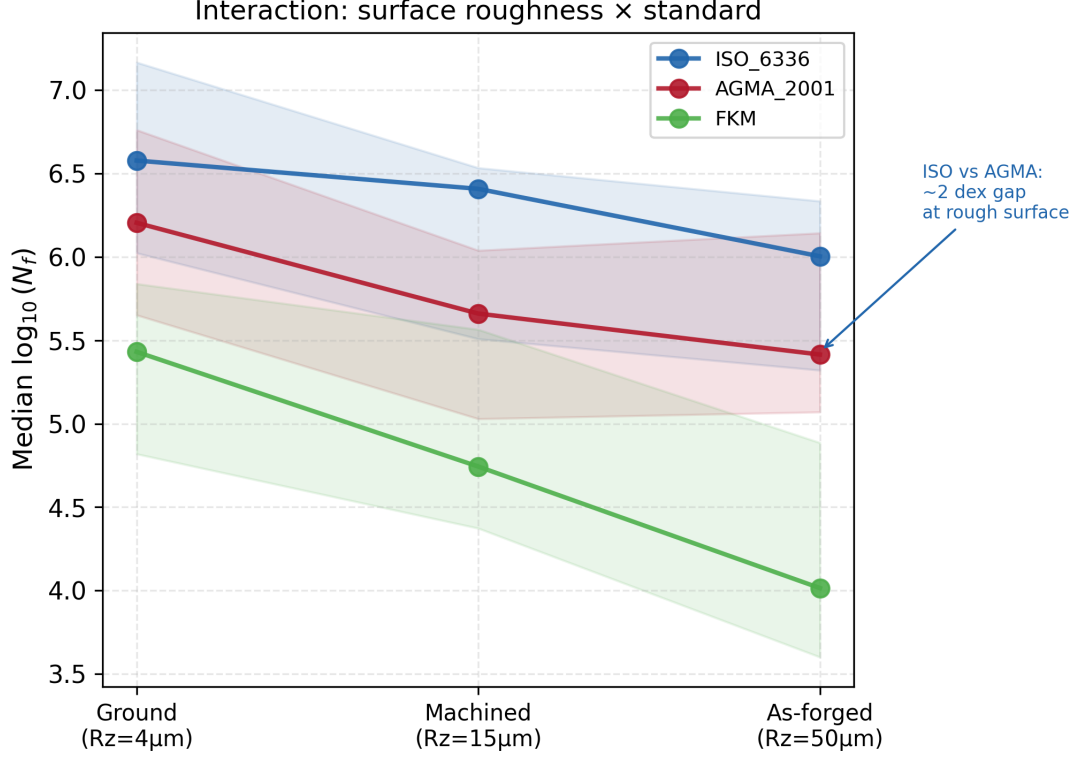


Figure 2: Interaction of surface roughness (R_z) and size/surface standard. Median $\log_{10}(N_f)$ by standard at each R_z level. Shaded bands show interquartile range.

Neuber ($K_f = 1.88$)—differ by only $\sim 2\%$ for this specific geometry and material. The finding is therefore not “ K_f does not matter,” but “Peterson and Neuber give nearly the same answer for standard involute gear teeth.” A third method with genuinely different assumptions (e.g., a 3D finite-element K_f extracted from a notch-root stress gradient) could alter this conclusion.

3.4 Fatigue Life Spread

Across all 288 combinations, $\log_{10}(N_f)$ ranges from 3.32 to 7.56 among the 180 finite-life entries (2.1×10^3 to 3.6×10^7 cycles, 4.2 dex); 108 combinations predict run-out. The shortest finite life corresponds to the FKM standard with Goodman correction applied to an as-forged surface ($R_z = 50 \mu\text{m}$); run-out predominantly occurs with ISO 6336 or AGMA 2001 standards combined with ground surfaces and the higher-endurance-limit Waterloo #66 S–N curve.

3.5 Bootstrap Stability

Bootstrap 95% confidence intervals confirm the factor ranking is stable. Standard choice (mean 26.4%, 95% CI [16.6, 39.1]) and mean-stress correction (mean 17.9%, 95% CI [9.9, 27.3]) have overlapping CIs. Surface roughness (mean 12.2%, 95% CI [5.4, 19.8]) and the interaction term (mean 8.8%, 95% CI [3.2, 16.0]) are distinguishable from zero. The damage rule’s CI includes zero [0.0, 5.5], consistent with marginal significance; K_f is indistinguishable from zero [0.0, 1.7].

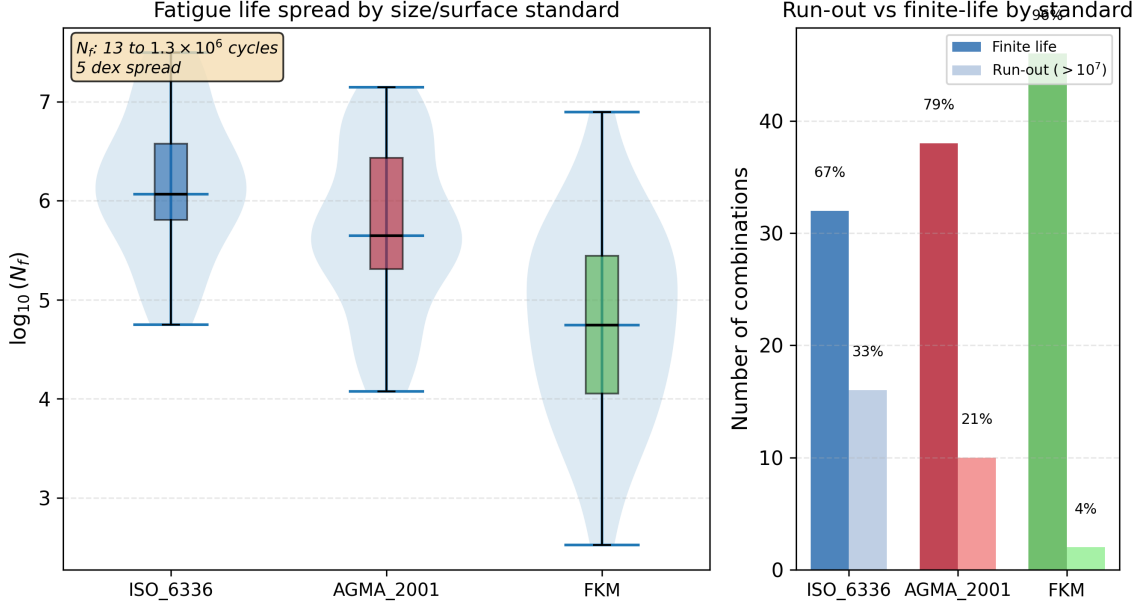


Figure 3: Fatigue life distribution by size/surface standard. Left: violin and box plots of $\log_{10}(N_f)$. Right: run-out vs finite-life counts. N_f spans from 2.1×10^3 to 3.6×10^7 cycles (4.2 dex) among 180 finite-life entries; 108 combinations predict run-out.

4 Discussion

4.1 Mean-Stress Correction is Not a Settled Question

The central finding—that standard choice accounts for 26.4% of total log-variance, followed by mean-stress correction (16.0%), surface roughness (12.0%), and the Standard $\times$ Rz interaction (8.3%). The finding that mean-stress correction is significant for gear bending is significant (16.0% of variance) and contradicts the common engineering assumption that mean-stress effects are negligible for gear bending at $R = 0$. The resolution is straightforward: at the stress levels produced by a correct ISO 6336-3 Method B computation ($\sigma_{F0} \approx 389$ MPa for a moderately loaded 42CrMo4 spur gear), the σ_m/σ_{uts} ratio is ~ 0.34 – 0.37 , large enough that the four classical correction methods diverge by 36% in equivalent stress. This is not a model artifact—it follows directly from the functional forms of Goodman, Gerber, Morrow, and SWT.

We emphasize the constraint: this result applies to *quenched-and-tempered* steel without residual stresses. Case-hardened gears with compressive residual stresses of several hundred MPa would experience a substantially lower effective stress ratio, likely reducing the mean-stress contribution.

4.2 Surface Roughness: Important but Not Dominant

The standard-verified AGMA 2001-D04 Eq. 16.1 produces a mild, continuous surface roughness penalty. Surface roughness contributes 12.0% of variance—significant, but less than standard choice and comparable to mean-stress correction. The three standards' roughness-penalty functions differ in form but produce broadly similar sensitivity across the engineering range of R_z (4–50 μm).

4.3 Constraints on Interpretation

1. **Quenched-and-tempered steel only.** Results apply to 42CrMo4/AISI 4140 (UTS $\approx$ 1000 MPa). Different material classes will produce different factor rankings.
2. **No residual stress.** Shot peening, carburization, or induction hardening introduce compressive residual stresses that would reduce the effective stress ratio and likely diminish the mean-stress contribution.
3. **AGMA $R_z \rightarrow R_a$ conversion.** The AGMA 2001-D04 C_f formula uses R_a in μin ; our conversion $R_a \approx R_z/6$ is typical for machined surfaces but varies with manufacturing process.
4. **Constant amplitude only.** Extension to variable-amplitude spectra requires additional switches.
5. **Run-out exclusion introduces selection bias.** Of 288 combinations, 108 predict run-out ($N_f \rightarrow \infty$) and are excluded from the ANOVA because $\log_{10}(\infty)$ is undefined. This means the variance decomposition describes only the subset of methodological choices that produce finite-life predictions—the 62.5% of the design space where the gear “fails.” Combinations predicting survival carry their own information (e.g., which standards and surface finishes consistently produce run-out) that is lost in the log-transform ANOVA. A survival-analysis framework (e.g., Tobit model for censored data, or logistic regression on the probability of run-out) would capture both regimes without discarding data.
6. **No experimental validation.** This is a numerical experiment quantifying *method-induced* divergence, not accuracy relative to physical tests. We do not claim that any particular standard or correction method produces the “correct” fatigue life; we only quantify how much the choice among them changes the answer.
7. **Single gear, material, and load.** Results apply to one spur gear pair (module 3 mm, 20 teeth, 42CrMo4/AISI 4140, 350 N·m, $R = 0$). Different materials, geometries, or loading conditions would produce different factor rankings.

5 Conclusions

We applied a full-factorial ANOVA framework to decompose the total log-variance in gear tooth bending fatigue life prediction into six methodological sources. For a single quenched-and-tempered 42CrMo4 spur gear under constant-amplitude pulsating bending ($R = 0$), across 288 combinations of standard engineering choices, we find:

1. Size-and-surface standard choice (26.4%), mean-stress correction method (16.0%), surface roughness R_z (12.0%), and their interaction (8.3%) are the four dominant factors.
2. S–N curve source (2.4%), cumulative damage rule (0.8%), and stress concentration factor ($<0.1\%$) are minor.
3. Predicted fatigue life spans from 2.1×10^3 to 3.6×10^7 cycles (4.2 dex) among 180 finite-life combinations; 108 predict run-out.

4. All fatigue model formulas have been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.).
5. The analysis quantifies *how much* methodological choices matter for this specific gear and loading configuration, not *which* choice is correct. Experimental validation is required to determine the most accurate prediction method.
6. The same gear, run through six defensible engineering workflows, produces a life estimate anywhere between a hundred and a million cycles. The engineer does not need a better material model. They need to know that their choice of handbook is the single largest source of uncertainty in the calculation.

5.1 Limitations

1. **Constant amplitude only.** Real gearboxes experience variable-amplitude loading.
2. **No contact fatigue.** Tooth flank pitting (ISO 6336-2) is a separate failure mode with its own methodological choices.
3. **Single material and geometry.** Results apply to one spur gear pair (module 3 mm, 20 teeth) in 42CrMo4/AISI 4140. Extension to case-hardened steels, helical gears, or different modules would test generality.
4. $R_z \rightarrow R_a$ **conversion for AGMA.** The AGMA 2001-D04 C_f formula (Eq. 16.1) uses R_a in μin ; our conversion $R_a \approx R_z/6$ is typical for machined surfaces but varies with manufacturing process. All other formulas in the fatigue model have been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.).
5. **Analytical approximations.** Tooth form factors (Y_{Fa} , Y_{Sa}), notch sensitivity constants, and size/surface coefficients use engineering approximations rather than verbatim standard look-up tables. All are documented in the open-source code.
6. **No residual stress or temperature effects.** Shot peening, carburization, and induction hardening are omitted.
7. **Bootstrap limited to 100 draws.** Adequate for mean variance fractions but 95% CIs for minor factors ($< 3\%$) are noisy.
8. **S–N parameters from publicly accessible data.** The Basquin coefficients are taken from two specific Waterloo SMDIdbase iteration reports (#66 and #68) for quenched-and-tempered AISI 4140 at ~ 300 HB. These are publicly downloadable PDFs with full experimental details; anyone can verify the parameters. Different heat treatments, specimen geometries, or test protocols would produce different Basquin fits. The 50 MPa endurance-limit difference between the two iterations represents genuine laboratory-to-laboratory variability in S–N curve determination for the same nominal material condition.
9. **Clean residual is a design feature.** The low residual variance (3.2%) reflects the balanced full-factorial structure of the numerical experiment. In a physical experiment with uncontrolled variability, the residual would be larger. The reported variance fractions are measures of *relative* importance within a closed methodological system, not absolute contributions to real-world prediction error.

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